



# RCC INSTITUTE OF INFORMATION TECHNOLOGY

## MAJOR PROJECT PRESENTATION

TITLE- AUTOMATED TRAIN DETECTION AND GATE CONTROL SYSTEM

MEMBERS – ANSHU RAJ, ADITYA SHARMA

MENTOR NAME – DR. APU MISTRY

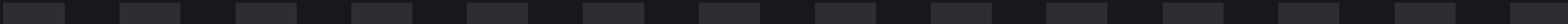
YEAR 2026



# Automated Train Detection and Gate Control System

An engineering solution combining vibration sensing, infrared detection, and ESP32-based automation to eliminate human error at level crossings.

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# Why Traditional Crossings Fall Short

In an increasingly urbanized world, traditional railway gate control methods frequently rely on manual operation or rudimentary automatic systems that lack sophistication. This project addresses these concerns through the sequential activation and interaction of different sensor technologies, creating a multi-layered defense against accidents.

122

Critical safety incidents recorded at crossings in India annually

85%

of accidents are attributed to human error and inadequate systems

# Multi-Sensor Detection Architecture

The system revolutionizes railway crossing safety through a sophisticated multi-sensor approach.



## Vibration Detection

SW420 sensor provides long-range detection of approaching trains.



## Sequential IR Sensing

IR sensors along the track provide sequential feedback.



## Escalating Alert

Buzzer provides audible warning with increasing intensity as the train approaches.



## Automated Control

Servo motor ensures smooth, automated gate operation based on confirmed detection.

## CRITICAL SAFETY INNOVATION

# Trapped Object Detection

A critical aspect of this system is its emphasis on internal safety. A dedicated internal sensor continuously monitors the area between the gates during the closure sequence.

In the event that a person or object becomes trapped during gate closure, this sensor immediately detects their presence, triggering the gate to re-open and allow for safe exit.



### 01 Continuous Monitoring

Internal sensor scans the gate area throughout closure.



### 02 Instant Detection

Identifies trapped persons or objects immediately.



### 03 Emergency Override

Triggers automatic gate re-opening for safe exit.



# Safety Insurance with Vibration Sensors

## Pre-Validation Devices

Vibration sensors placed before the IR sensors act as an early, independent confirmation stage.

## Fewer False Triggers

Reduces false triggering of IR sensors caused by humans, animals, or environmental disturbances such as fog and rain.

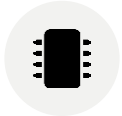
## Layered Reliability

This layered detection approach improves system reliability, ensures accurate gate operation, and prevents premature gate closure.

## Validation Pipeline



# Hardware Architecture & Components



## ESP32 Microcontroller

Central processing unit managing sensor inputs and output controls with enhanced processing power.



## SW420 Vibration Sensor

Detects train-induced mechanical vibrations from long distance, providing early trigger.



## IR Sensor Array

Placed at calculated intervals along the track for sequential detection.



## SG90 Servo Motor

Physically opens and closes the gate with precise control.



## Auditory Buzzer

Provides escalating audio signals indicating train proximity.

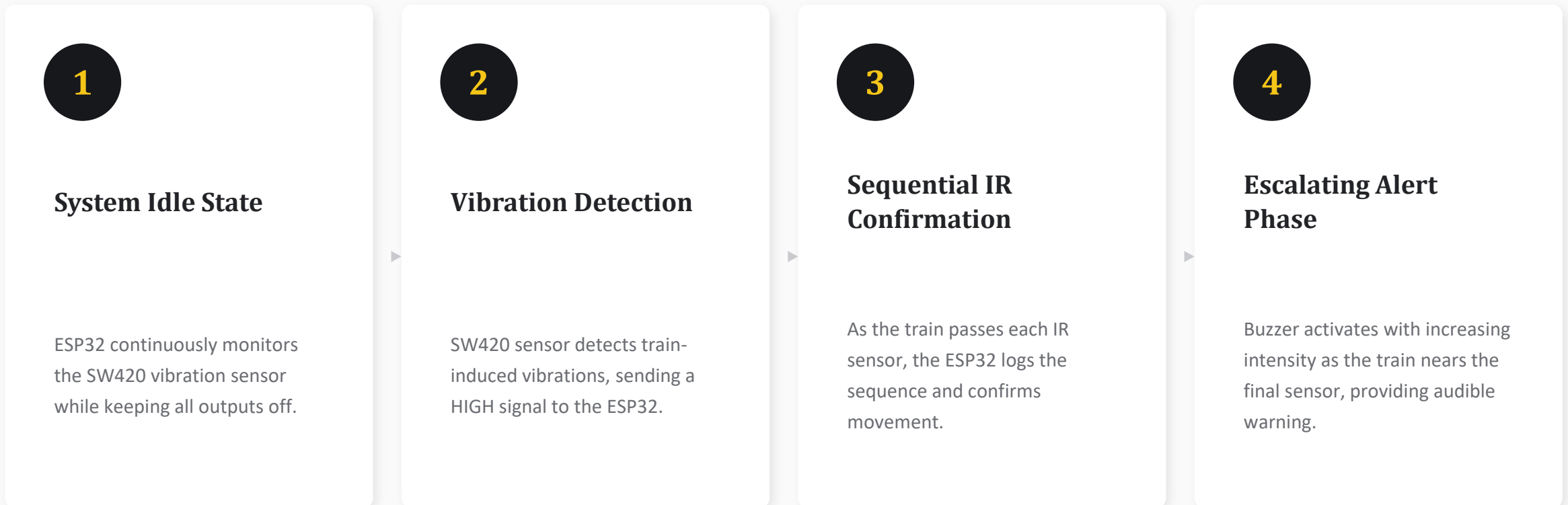


## Internal Detection Sensor

Monitors the area between gates for trapped objects, enabling safe exit protocols.

# System Operational Flow — Approach

The operational flow is designed to be sequential, responsive, and fail-safe, ensuring maximum safety.





## System Operational Flow — Closure & Reset

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### Gate Closure Operation

When no obstruction is detected, the ESP32 commands the servo motor to close the gate.

6

### Safety Monitoring

The internal sensor continuously scans for trapped objects. If detected, the system reopens the gate immediately for safe exit.

7

### Train Departure & Reset

After the vibration/IR sensors confirm the train has passed, the gate reopens safely and the system resets.

## SOFTWARE

# Software Implementation

The control software is written using C++ in the Arduino IDE.



### Interrupt Routines

Handle real-time events from the vibration trigger for immediate system response.



### Finite State Machine (FSM) Logic

Manages different states including idle, warning, closing, waiting, and opening phases.



### PWM Signal Generation

Provides precise gate movement control via the servo motor with smooth operation.



### Digital Input Monitoring

Tracks sensor states continuously and executes conditional logic for safety.

## VALIDATION

# Comprehensive Testing Protocol

Every stage of the system is validated end-to-end before field deployment.



### Sensor Calibration

Tune vibration and IR thresholds for reliable, repeatable triggering.



### Response Time Measurement

Measure latency from detection to gate action.



### Gate Operation Trials

Repeated open/close cycles under varied conditions.



### Alert Feedback Test

Confirm buzzer escalation timing and audibility.



### Safety Validation

Verify trapped-object detection and emergency override.

**Thank You!**